

Daniel Peelen

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EDUCATION

University of Pittsburgh, School of Computing and Information

Pittsburgh, PA

Bachelor of Science in Computer Science, Minor in Hungarian

Expected May 2027

- **GPA:** 3.9/4.0 (Dean's List)
- **Coursework:** Intro to Deep Learning, Intro to ML, Data Structures & Algorithms, Probability Theory

WORK EXPERIENCE

Software Engineering Intern

Expected May 2026 – August 2026

Microsoft

Redmond, WA

- Selected for Microsoft Word AI team to ship LLM features for document creation and consumption workflows.

Machine Learning Engineering Intern

January 2026 – April 2026

Synopsys

Pittsburgh, PA

- Improved RAG coding agent accuracy from 40% to 82% by designing a semantic clustering memory system.
- Built a local multi-turn eval harness with bucket visualization to benchmark memory on Ansys test infrastructure.
- Shipped memory solution to production by porting to Go and containerizing with feature-flag support.

Machine Learning Engineer Intern

June 2025 – August 2025

Thursday

Pittsburgh, PA

- Architected end-to-end production ML pipeline delivering personalized fashion recommendations to 10K MAUs.
- Optimized inference endpoints with Redis caching and batching; achieved <200ms latency on predictions.
- Built distributed Kafka scraping infrastructure using 100+ sites to fuel CV model training pipelines at scale.

Software Engineer Intern

May 2025 – August 2025

PNC Financial Services Inc.

Pittsburgh, PA

- Built full-stack housing analytics app (Spring Boot, Angular) processing 1M+ records; raised test coverage by 40%.

PROJECTS

DannyTorch | *Python, NumPy*

January 2026 - April 2026

- Built an ML framework from scratch with NumPy; built full reverse-mode autograd over arbitrary compute graphs.
- Coded 10+ transformer parts including MHA, RoPE, LayerNorm, MLP with Dropout, Adam and more.
- Validated gradient correctness by training a character-level LM on n-gram tasks for 500 epochs on a Colab A100.

Continuum Memory Transformer | *PyTorch, NumPy, CUDA*

November 2025 – December 2025

- Built 36M-parameter transformer with RoPE, GQA, and a hierarchical continuum memory module from scratch.
- Tested 4 variants on WikiText-2; quantified memory tradeoff of +2.8% perplexity, -30% throughput at scale.

AnkiMate | *Python, Flan-T5, PyTorch, Flask, React*

May 2025

- Fine-tuned Flan-T5 on self-curated 10K-row dataset to make cards from PDF/PPT, reduced validation loss 58%.
- Designed and deployed a Flask backend for flashcard generation; used React with Tailwind CSS for a responsive UI.

Mantis | *Python, YOLOv8, OpenCV, GCP, Docker*

October 2025

- **2nd Place, CalHacks 2025 (600+ teams).** Real-time CV system; 89% accuracy on shoplifting and weapons.

SKILLS

Languages: Python, C/C++, Go, Java, JavaScript, TypeScript, HTML/CSS, OCaml, SQL

Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, NumPy, Pandas, OpenCV

Infra & Tools: Docker, Kafka, Redis, GCP, AWS, Git, GitHub Actions, Linux

Concepts: Transformers, Autograd, RAG, Distributed Training, LLM Fine-tuning, Computer Vision, RLHF

EXTRACURRICULARS

Director, SteelHacks

January 2025 – September 2025

- Pittsburgh's Largest Hackathon — 600+ attendees, \$50K+ raised, sponsors incl. Google, Amazon, PNC

CITIZENSHIP

USA, Hungary (*dual; authorized to work in US or EU without sponsorship*)